

Course: B. Pharmacy

Subject Name: Physical Pharmaceutics I

Max Marks: 75

Sem: III

Subject Code: BP302T

Duration: 3 Hrs

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Differentiate between solid, liquid and gas.
- ii) State Raoult's law.
- iii) What is eutectic mixture?
- iv) Define interfacial tension and surface tension.
- v) Write application of buffer in pharmacy.
- vi) What is critical solution temperature? Give example.
- vii) Write Henderson Hasselbalch equation for acid and base.
- viii) Define polymorphism along with example.
- ix) Draw neat labeled diagram of HLB scale.
- x) Give importance of protein binding.

Q.2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) What is surface tension and surface free energy? Explain in detail drop count method for determination of surface tension.
- ii) Define complexation and explain different classes of complexes in detail
- iii) State phase rule and explain in detail phenol-water system with neat diagram

Q.3. Short Answers (Answer 7 out of 9)

(7 x 05) = 35

- i) Explain crystalline, amorphous & polymorphism nature in solid substances
- ii) Explain different factors affecting on solubility of drug.
- iii) Enlist different physicochemical properties. Explain refractive index and optical rotation.
- iv) Write a detail note on liquid crystals.
- v) What is partition coefficient? Write a note on it along with its formula.
- vi) Define isotonic and hypertonic solution. Describe Sorensen's pH scale.
- vii) Explain any one method to determine pH of solution.
- viii) Summarize concept of solubilization and detergency.
- ix) Explain any one method of adjusting tonicity.